Classical Systems Biology

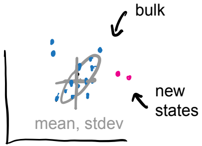
MCSB Bootcamp — Mathematical and Computational Track

Jun Allard

The goals, challenges, systems, tools and approaches of today

Data and analytics

Keystone **goal**: discover the hidden — new states, cell types, disease classes, ...



Keystone **challenge**: biological data is not just big, it is **high-dimensional**.

- 30000 genes, 1000 cells
- 10^4 microbial species, 20 patients

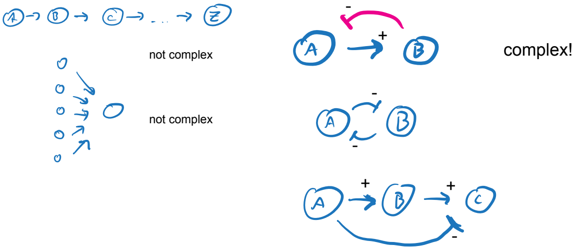
Keystone **tool**: clustering, regression, dimension-reduction ... coding.

Models and mechanism

Keystone **goal**: **prediction** — "what would happen if..."

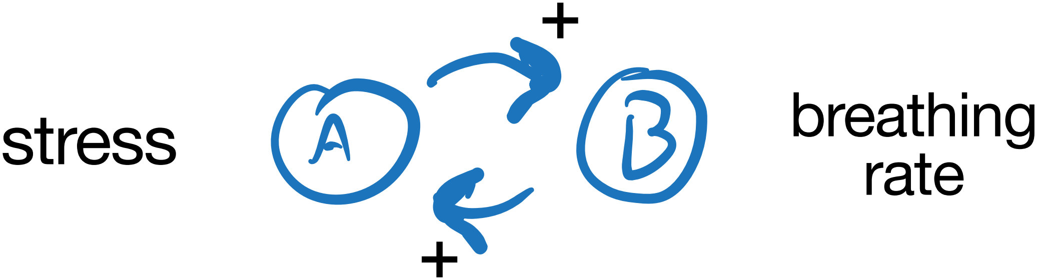
if A is increased, then...

Keystone **challenge**: biological systems are **complex**, not just big.



Keystone **tool**: dynamical systems, control theory, (the differential equation,) ...

Stress and breathing rate



Complex systems and “systems thinking”

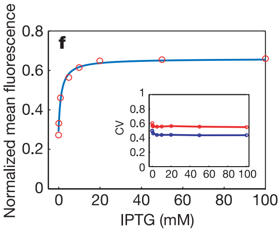
Early synthetic biology

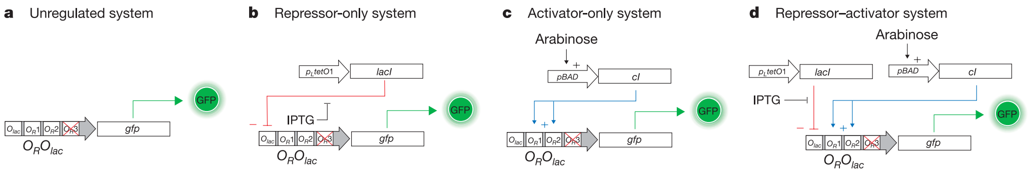
nature Vol 439|16 February 2006|doi:10.1038/nature04473

LETTERS

A bottom-up approach to gene regulation

Nicholas J. Guido[1]*, Xiao Wang[2]*, David Adalsteinsson[3], David McMillen[5], Jeff Hasty[6], Charles R. Cantor[1], Timothy C. Elston[4] & J. J. Collins[1]





"Computation drove the field of synthetic biology in its very early days."

Collins, 2020

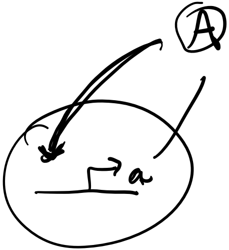
"We didn't have those data sets. We didn't have a wet lab. We didn't have the capability. We had to rely on mathematical modeling."

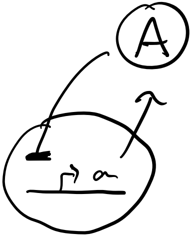
paraphrasing Collins, 2015

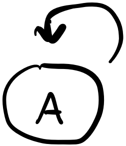
Complex biological systems and "systems thinking"

 Activation

 Repression







Positive feedback

With positive feedback, the curve showing steady-state protein as a function of activator TF should move up.

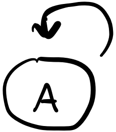
Does the curve move up by:

- shifting the EC50 to the left?
- moving up at high-C?
- moving up at low-C?
- a combination of all three?

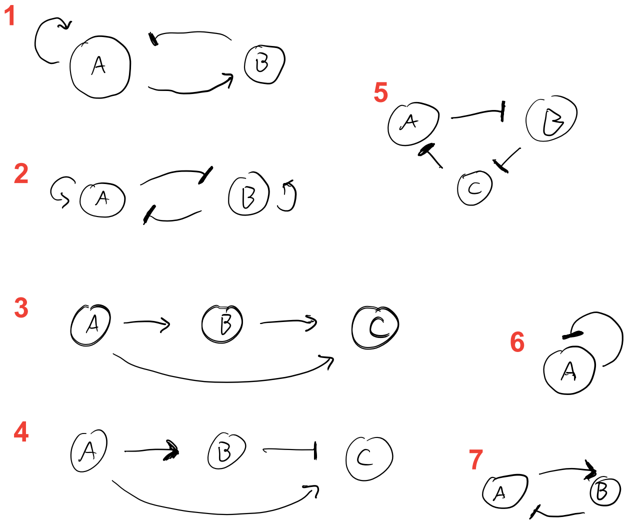
Seven circuits

Activation

Repression

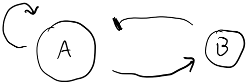


Positive feedback



In-class activity: matching

1



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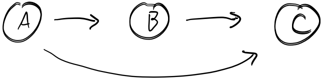

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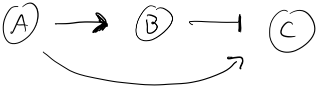

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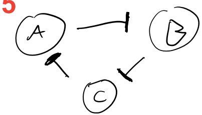

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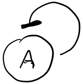

7

Match each circuit to its name.

- Direct negative feedback
- Indirect negative feedback
- Double-negative feedback
- Fast positive and slow negative feedback
- Incoherent feedforward
- Coherent feedforward
- The Repressilator